

Learning to zoom: Exploiting mixed-scale contextual information for object detection

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ARTICLE INFO

Keywords:

Deep neural networks
Object detection
Mixed-scale
Contextual information

ABSTRACT

With the development of deep neural networks, object detection has made a significant leap. However, apart from intrinsic intra and inter class differences, the objects in real-world scenarios may encounter diverse scales, blurred appearance or even severe occlusion. Inspired by the human behavior of looking at blurred images, *i.e.*, zooming in/out, we propose a Mixed-Scale Network (dubbed for MSNet) to tackle these issues. Specifically, MSNet employs the zoom strategy to exploit mixed-scale contextual information, which fully unleashes the representation ability of deep neural networks. Firstly, the global feature aggregation module and global feature enhancement module aims at aggregating mixed-scale features from the global perspective, which learns the transform offset of pixels to align the higher-res features contextually. Moreover, the local feature aggregation module enriches the instance-level feature by adaptively aligning the features of different receptive fields. Extensive experiments on MS COCO demonstrated the effectiveness of MSNet, yielding significant improvements of 0.7–2.8 points in AP_{box} over baselines when paired with different detectors, backbones, and schedules. In addition, for pixel-level prediction tasks including semantic segmentation and instance segmentation, the proposed method gains consistent improvements of 0.6–2.7 points in mIoU/ AP_{mask} over baselines, which substantiates the effectiveness of MSNet further.

1. Introduction

Object detection is a fundamental topic in the field of computer vision, which aims at categorizing and localizing objects existing in an image simultaneously. It has a wide variety of applications in the real-world, such as autonomous driving (Cai et al., 2021), anomaly detection (de Paula Monteiro, Lozada, Mendieta, Loja, & Bastos Filho, 2022), and video surveillance (Doshi & Yilmaz, 2022). Due to the advancement of deep convolutional neural networks (CNNs) (He, Zhang, Ren, & Sun, 2016; Ji, Du et al., 2020; Ji, Wen et al., 2020; Simonyan & Zisserman, 2015; Xie, Girshick, Dollár, Tu, & He, 2017; Xu, Liu, Cao, et al., 2021), in recent years, object detection has undergone remarkable improvements in both accuracy and speed. Typically, mainstream CNN-based detectors can be roughly categorized into two types: one-stage and two-stage detectors. The two-stage detectors mainly consist of two steps. They first generate a sparse set of candidate regions using a region proposal method, such as selective search (Uijlings, Van De Sande, Gevers, & Smeulders, 2013) and region proposal network (RPN) (Ren, He, Girshick, & Sun, 2017), and then compute fine-tuned predictions

based upon candidate regions. This scheme achieves state-of-the-art performance on various challenging object detection benchmarks, *e.g.*, PASCAL VOC (Everingham et al., 2015), MS COCO (Lin et al., 2014). On the other hand, the one-stage detectors utilize well-designed architecture to predict the classification and regression of interest directly, without the procedure of generating candidate regions. In general, the one-stage detectors have faster detection speed but usually lag behind the two-stage detectors in terms of detection accuracy.

However, there are still some problems in real-world scenarios due to the diversity of scale, fuzziness, and occlusion of appearance, which significantly hinder the accuracy of the detector. Target mitigating these problems, some pyramid-based techniques have been proposed, including Image Pyramid and Feature Pyramid, which excavate discriminative contextual cues across instances through a hierarchical structure. Both group methods share the same motivation, that is, a desired model should possess different receptive fields to meet variant scale objects. (1). The image pyramid methods (Singh & Davis, 2018;

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<https://doi.org/10.1016/j.eswa.2024.125871>

Received 29 June 2023; Received in revised form 6 June 2024; Accepted 19 November 2024

Available online 26 November 2024

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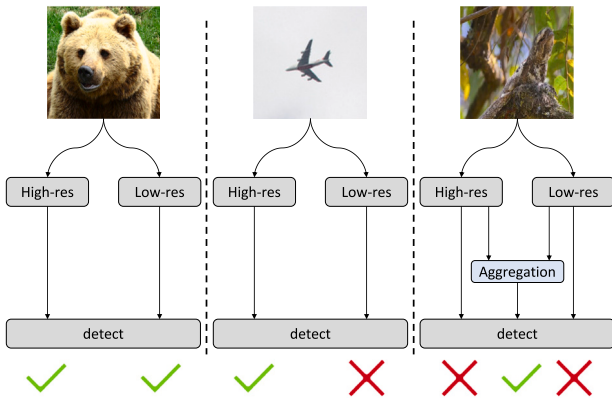


Fig. 1. The prediction results of a pre-trained detector for samples under different resolutions.

Singh, Najibi, & Davis, 2018) take multi-scale images as input and perform object detection at each scale individually. Although they bring an extra computational burden, the image pyramid methods boost the representation ability of model and prevent objects from overfitting to a certain scale. (2). The feature pyramid methods use different levels of features to detect objects of different scales, which is computation-friendly. Each feature map in the pyramid is built on a certain level of the backbone, which is used for detecting objects in a specific range of size. For example, FPN (Lin, Dollár et al., 2017) integrates adjacent features through lateral connection and top-down pathway to build a feature pyramid with strong semantics on all scales. PAFPN (Liu, Qi, Qin, Shi, & Jia, 2018) further improves the localization capability of the feature pyramid by adding additional bottom-up paths. Libra R-CNN (Pang et al., 2019) exploits the features from all levels to generate more balanced semantical features. However, owing to the dependence on fixed scale, the representation ability of above methods is restricted.

From the perspective of real life, human beings may zoom the image to seek more contextual clues when they cannot accurately identify objects. This particular human behavior inspires us to detect objects by imitating the zoom strategy. As shown in Fig. 1, for a clear graphical explanation, we provide the image resolution required to give correct prediction with the pre-trained detector. In practice, “easy” samples, e.g., bear with obvious foreground, can be correctly classified at both low and high resolution, while “difficult” samples, e.g., such as bird with entangled foreground and background, need to be accurately predicted under the aggregation of high and low resolution. Recently, ZoomNet (Pang, Zhao, Xiang, Zhang, & Lu, 2022) employs the zoom strategy to learn the mixed-scale global semantics for camouflaged object detection, and achieves the great success. However, since there is a huge difference between camouflage object detection and generic object detection. The camouflaged object detection is a pixel-level prediction task, which segments hidden objects that blend into the surrounding environment. The generic object detection is an instance-level prediction task, which mainly identifies objects in images and gives specific category and location information. Directly applying ZoomNet to object detection may impose a huge computational burden.

In this paper, to achieve the trade-off between latency and accuracy, we propose a simple yet efficient structure to fully unleash the representation ability of deep neural networks, named MSNet. Concretely, the proposed method mimics the zooming strategy to form an expressive representation of the original high-res detection model from both global and local perspectives. Firstly, we input the mixed-scale images into the backbone to extract the corresponding features and adaptively tweak the feature maps extracted from the backbone at each stage by the designed global feature aggregation module (GFAM). Then, we propose a global feature enhancement module (GFEM) to zoom the inherent highest-level feature and refine its representation, which alleviates

the problem of information loss caused by lateral connection. Finally, the local feature aggregation module (LFAM) is designed to further refine the local feature for each instance via adaptively aggregating spatial-wise contexts of different scales.

The main contributions of this paper are summarized in the following aspects:

- We propose a novel Mixed-Scale Network for object detection, where the global feature aggregation module adaptively tweaks the feature maps extracted from the backbone at each stage; the global feature enhancement module is proposed to alleviate the problem of information loss and enrich the representation of highest-level features; the local feature aggregation is designed to refine instance presentation via adaptively aggregating feature of different scale.
- MSNet can be flexibly plugged into the existing bottom-up backbone network and improve its performance significantly, which makes a decent trade-off between latency and accuracy.
- Extensive experiments conducted on the MS COCO benchmark show that MSNet gains consistent performance improvement of **0.7%–2.8%** AP_{box} over existing methods. Moreover, MSNet outperforms baseline by a large margin of **0.6%–2.7%** $AP_{mask}/mIoU$ in pixel-level prediction tasks including semantic segmentation and instance segmentation.

The remainder of this paper is organized as follows. Section 2 gives a short overview of the related works. In Section 3, the proposed mixed-scale network is presented in detail. Finally, we conduct extensive experiments and summarize the entire paper in Sections 4 and 5, respectively.

2. Related work

2.1. Object detection

In recent years, thanks to the rapid development of deep learning techniques, remarkable progress has been achieved in object detection. The mainstream object detection algorithms are roughly divided into two categories: two-stage and one-stage. The paradigm of two-stage predicts the object proposals first and then calculates the refined prediction based on these proposals. R-CNN (Girshick, Donahue, Darrell, & Malik, 2014) is a milestone algorithm of this strand research, which shows that the convolutional neural network can significantly improve detection performance. Later, SPP-Net (He, Zhang, Ren, & Sun, 2015) and Fast R-CNN (Girshick, 2015) propose the SPP pooling or ROI pooling to reduce the redundant RoI feature computation burden, which significantly speeds up the detection process. Faster R-CNN (Ren et al., 2017) introduces the Region Proposal Network (RPN) instead of selective search to cut the computational cost further. After that, a broad range of follower (Cai & Vasconcelos, 2018; Chen, Pang et al., 2019; He, Gkioxari, Dollár, & Girshick, 2017; Zhang, Chang, Ma, Wang and Chen, 2020; Zhu et al., 2017) are proposed to advance Faster R-CNN (Ren et al., 2017) from different aspects. For example, Mask R-CNN (He et al., 2017) develops a mask branch to Faster R-CNN, which improves detection performance through multi-task learning. Cascade R-CNN (Cai & Vasconcelos, 2018) devises a multi-stage architecture to boost the detection results in a cascade manner.

With the increasing demand for speed and accuracy, the one-stage methods (Lin, Goyal, Girshick, He and Dollár, 2017; Liu et al., 2016; Redmon, Divvala, Girshick, & Farhadi, 2016; Redmon & Farhadi, 2017, 2018; Zhang, Wen, Lei and Li, 2020) shine under the spotlight, which directly predict the classification and localization of each object without refinement process. As one of the first attempts, OverFeat (Sermanet et al., 2014) constructs multi-scale windows on the shared convolutional feature maps to detect objects. The family of YOLOs (Redmon et al., 2016; Redmon & Farhadi, 2017, 2018) simplifies object detection as a regression problem, which directly predicts the bounding boxes

and class probabilities through a single feed-forward network. Later, SSD (Liu et al., 2016) utilizes multiple feature maps at different scales to further boost the detection performance. However, compared with the two-stage detectors, the early work has faster speed, but tailed in the accuracy. RetinaNet (Lin, Goyal et al., 2017) introduces the Focal Loss to alleviate the issue of foreground-background class imbalance and makes it possible for the one-stage detectors to exceed the two-stage detector. More recently, the efforts have been focused on the emerging anchor-free based one-stage methods (Duan et al., 2019; Feng, Zhong, Gao, Scott, & Huang, 2021; Law & Deng, 2020; Tian, Shen, Chen, & He, 2019; Zand, Etemad, & Greenspan, 2022; Zhou, Zhuo, & Krähenbühl, 2019), whose goal is to get rid of the limitation of the pre-defined sliding windows or proposals.

2.2. Pyramid network

Pyramid network has been proven an effective component to extract contextual cues in object detection. The existing pyramid network can fall into two categories. The commonly-used method is image pyramid methods, which takes multi-scale images as input and carried out feature extraction and object detection for each scale respectively. Therefore, objects with various scales can be detected under proper receptive fields. SNIP (Singh & Davis, 2018) and SNIPER (Singh et al., 2018) are the representative image pyramid methods. Nevertheless, its high computation costs pose a challenge for practical application.

Another solution is the feature pyramid. The inherent multi-scale features of the backbone network are exploited to detect objects with various scales on distinct layers. As for the powerful ability to aggregate context information, feature pyramid architecture is widely utilized for improving the performance of object detection. SSD (Liu et al., 2016) is a pioneer to perform object detection on the pyramidal features. FPN (Lin, Dollár et al., 2017) propagates high-level semantic information to the shallow level by a top-down pathway with lateral connections at all scales. Later, many follow-up works (Chen, Zhao et al., 2023; Ghiasi, Lin, & Le, 2019; Guo, Fan, Zhang, Xiang, & Pan, 2020; Liu et al., 2018; Pang et al., 2019; Wang et al., 2019; Zhu, Lee, Cai, Yu, & Chen, 2022) are proposed based on FPN architecture. For example, PAFPN (Liu et al., 2018) further improves the localization capability of the feature pyramid by adding additional bottom-up paths. Libra R-CNN (Pang et al., 2019) aims to generate more balanced semantic features by using features at all levels. Recently, the neural architecture search (NAS), which utilizes the reinforcement learning strategy to automatically derive the optimal neural network architecture, has led to great success in classification. Ghiasi et al. (2019) first utilizes NAS to explore the pyramid network with the best performance, and proposes a powerful pyramid network called NAS-FPN. However, the network derived by NAS is extremely complex and lacks a reasonable explanation. In addition, it brings a lot of parameters and computational burden, which hinders practical application.

Note that our method is obviously different from the image pyramid. Although mixed-scale images are introduced as input, we do not carry out detection for each scale respectively but divide the mixed-scale into the auxiliary scale and main scale (i.e., original input scale).

2.3. Attention mechanism

Motivated by the human visual system, there are many efforts dedicated to attention mechanisms in the literature, which can effectively guide the network to focus on relevant regions of an image and suppress irrelevant parts. SEBlock (Hu, Shen, & Sun, 2018) recalibrates the feature map by using the signals aggregated from the whole feature map. In this way, spatial-irrelevant information is learned to capture channel-wise dependencies. Inspired by SEBlock, CBAM (Woo, Park, Lee, & Kweon, 2018) extends the attention mechanism to the spatial axis. CBAM is composed of two modules, i.e., a channel-wise attention

block and a spatial attention block, which guide the network to distinguish the informative channel and spatial regions. BAM (Park, Woo, Lee, & Kweon, 2018) also explores the association between channel and spatial attention. A parallel structure was adopted to realize the interaction between channel attention and spatial attention. Considering that self-attention mechanism has shown great advances in the field of natural language processing, Wang, Girshick, Gupta, and He (2018) took the lead in introducing self-attention to computer vision and presented a novel non-local block. Non-local adopts self-attention mechanism to characterize remote dependency, which aggregate information from all positions. In order to reduce redundant computations, GCBLOCK (Cao, Xu, Lin, Wei, & Hu, 2019) combines the advantages of Non-local (Wang et al., 2018) and SEBlock (Hu et al., 2018) to form rich global context information. Recently, attention mechanism is introduced into the computer vision community and is widely applied in various visual tasks, e.g., image classification (Hu, Miao, & Pedrycz, 2022), image caption (Wang, Shen, & Ji, 2022), and visual question answering (Chen, Han, & Chang, 2022).

3. Methodology

This section gives the detailed description and rationale behind the Mixed-Scale Network. The overall architecture is shown in Fig. 2, which is mainly formed by three modules: Global Feature Aggregation Module, Global Feature Enhancement Module, and Local Feature Aggregation Module.

3.1. Global feature aggregation module

Inspired by the above observation, i.e., zooming behavior, we believe that different resolutions of an image offer multi-granularity information. High-res can provide more detailed clues, such as edges and textures while low-res contains the approximate content of the image. Thus, we utilize the auxiliary scale (low-res branch) as a supplement to the original main scale (high-res branch), in which the auxiliary scale is obtained by “zooming” the main scale. After obtaining the mixed-scale images, we input them into the backbone network to extract the corresponding features. To fuse these mixed-scale features, we design a global feature aggregation module (GFAM) to enrich the original feature in an adaptive aggregation manner. After that, we realize the supplement of auxiliary-branch information to main-branch information.

The details of the GFAM can be seen in Fig. 3. Given an image $I \in \mathbb{R}^{H \times W \times 3}$, we first obtain its corresponding low-res image $I^p \in \mathbb{R}^{kH \times kW \times 3}$ by averaging pooling operation (k is the downsampling ratio). Then, the backbone network extracts features of several resolutions at different stages from the input images respectively. After that, GFAM is applied to adaptively aggregate high-res and low-res features. For i_{th} stage of backbone, $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ and $F_i^p \in \mathbb{R}^{H_i^p \times W_i^p \times C_i}$ are the corresponding features of the inputs. It should be noted that $H_i^p = kH_i$ and $W_i^p = kW_i$. The whole adaptive alignment process can be summarized as follows:

Firstly, we upsample low-res feature $F_i^p \rightarrow Up(F_i^p)$ by bilinear interpolation to match spatial scale. Element-wise addition operation is applied to aggregate features F_i and $Up(F_i^p)$ to produce their global semantic representation $G_i \in \mathbb{R}^{H_i \times W_i \times C_i}$.

Then, in order to decrease computational complexity, we use average pooling followed by maximum pooling to obtain contextual features $G_i^{Avg} \in \mathbb{R}^{H_i \times W_i \times 1}$ and $G_i^{Max} \in \mathbb{R}^{H_i \times W_i \times 1}$ along spatial dimension. After that, we can obtain the refined global representation $G_i^{refine} \in \mathbb{R}^{H_i \times W_i \times 2}$ by concatenating two contextual features along the channel dimension, which is formulated as follows:

$$G_i^{Avg}(m, n) = \frac{1}{C} \sum_{d=1}^C G_i(m, n, d) \quad (1)$$

$$G_i^{Max}(m, n) = \text{Max}(G_i(m, n, 1), \dots, G_i(m, n, C)) \quad (2)$$

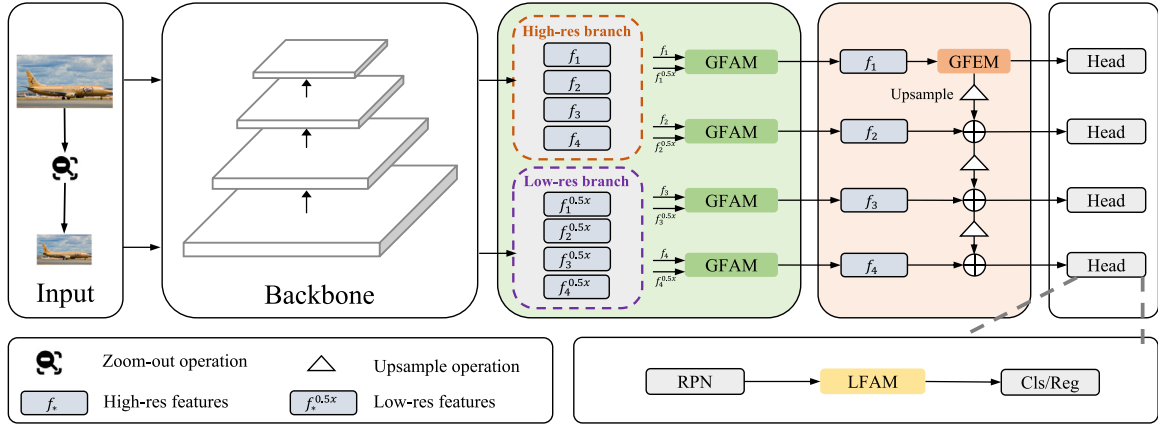


Fig. 2. The overall framework of MSNet. The proposed model consists of three sub-components, including Global Feature Aggregation Module (GFAM), Global Feature Enhancement Module (GFEM), and Local Feature Aggregation Module (LFAM). RPN: Region Proposal Network. Cls/Reg: Classification/Regression.

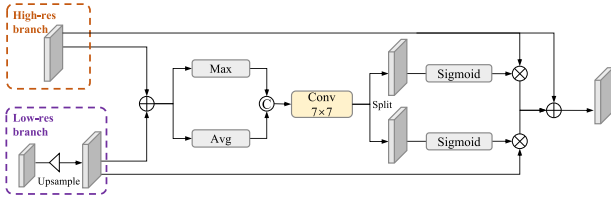


Fig. 3. The overall framework of GFAM. Max: Maximum Pooling operation. Avg: Average Pooling operation.

$$G_i^{refine} = \text{Concat}(G_i^{Avg}, G_i^{Max}) \quad (3)$$

where $\text{Concat}(\cdot)$ and $\text{Max}(\cdot)$ represents the concatenation and maximum operation, respectively. As in SENet and CBAM, average pooling and maximum pooling are the most common pooling operations. Inspired by CBAM, we jointly adopt maximum pooling and average pooling operations to obtain a more comprehensive global representation.

Finally, to highlight the importance of relevant features, we use sigmoid function to allocate different weights between resolution features, see Eq. (4). In this way, we can adaptively align features of different resolutions by the weighed sum of each input.

$$W_i = \delta(\text{Conv}(G_i^{refine})) \quad (4)$$

$$X_i = W_i^1 \cdot F_i + W_i^2 \cdot U p(F_i^p) \quad (5)$$

$$\delta(x) = \frac{1}{1 + \exp(-x)} \quad (6)$$

where W_i represents the weight of features with different resolutions, $\delta(\cdot)$ denotes sigmoid function, $[\cdot]$ denotes element-wise multiplication operation, $+$ denotes element-wise addition operation.

3.2. Global feature enhancement module

Following the common protocol of FPN (Lin, Dollár et al., 2017), the lateral connection is applied to fuse multi-scale features in a top-to-down fashion. However, it brings the side product of reduction in the channel dimension, which inevitably causes information loss across scales. Generally, the highest-level feature usually is formed by thousands of channels, which is important to characterize semantic representation. To alleviate this issue, similar to the idea in GFAM, we propose a global feature enhancement module (GFEM) for stable and discriminative representation. Concretely, GFEM introduces scale-specific semantic information to supplement the original features at

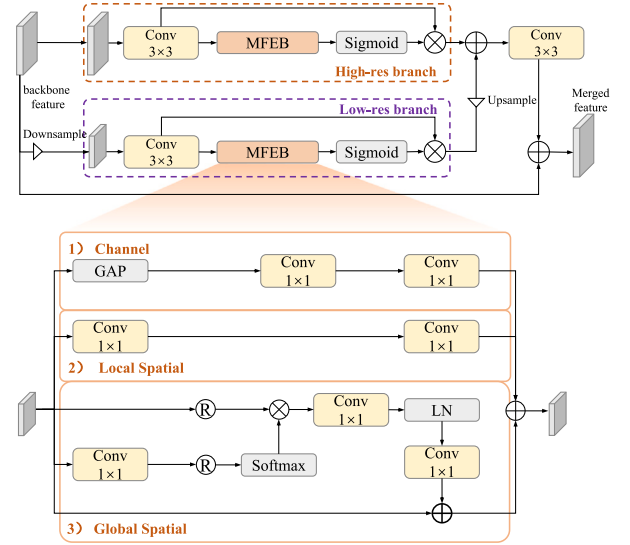


Fig. 4. The overall framework of GFEM. MFEB: Multi-granularity Feature Enhancement Block. GAP: Global Average Pooling operation. R: Reshape operation. LN: LayerNorm operation.

the specific stage. Fig. 4 illustrates the flow-chart of global feature enhancement module.

Specifically, suppose that the input feature is $X_i \in \mathbb{R}^{H_i \times W_i \times C_i}$. First, we can obtain the pooled feature $X_i^p \in \mathbb{R}^{dH_i \times dW_i \times C_i}$ by performing an average pooling operation on the original feature X_i , and downsampling ratio is d . In this way, the whole module is divided into two parallel branches, i.e., high-res branch and low-res branch. Then, we obtain the refined features $\hat{X}_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ and $\hat{X}_i^p \in \mathbb{R}^{dH_i \times dW_i \times C_i}$ by using the 3×3 convolutional layer on each branch. After that, the features are sent to the Multi-granularity Feature Enhancement Block (MFEB) to further mine informative contextual features from different aspects. Taking the high-res branch as an example, the MFEB is divided into three levels:

Channel. In light of the fact that channels characterize rich feature information, we introduce the channel attention module to exploit the inter-channel information. To be concrete, a global average pooling layer followed by two convolutional layers with kernel 1×1 is used to compress spatial information and calculate the correlations between channels. Thus, channel-specific information $F_c \in \mathbb{R}^{1 \times 1 \times C_i}$ is calculated as:

$$F_c = H(\text{GAP}(\hat{X}_i)) \quad (7)$$

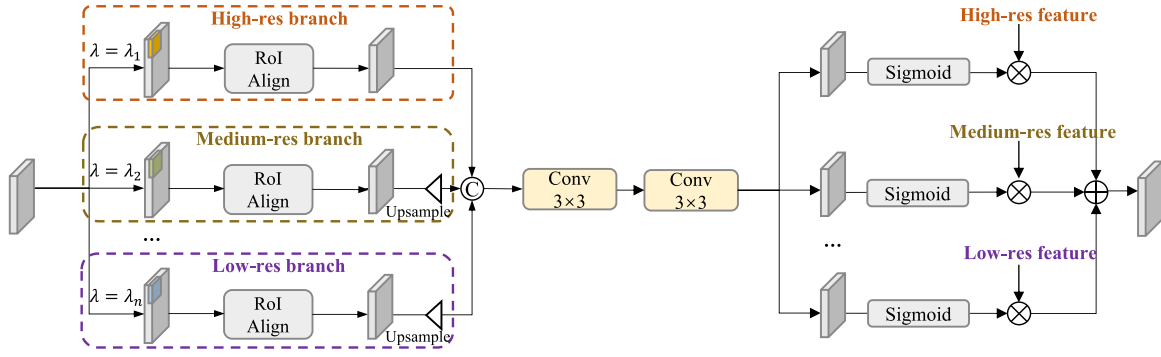


Fig. 5. The overall framework of LFAM.

where $\mathcal{H}$ indicates the two convolutional operations, and $\text{GAP}(\cdot)$ represents the global average pooling operation.

Local Spatial. The spatial-wise features usually present detailed information about the image. Modeling a local spatial attention module can effectively activate the regions of interest while suppressing irrelevant ones. Hence, we use two 1×1 convolutional layers to mine local spatial semantic information $\mathcal{F}_{ls} \in \mathbb{R}^{H_i \times W_i \times C_i}$ in this paper.

$$\mathcal{F}_{ls} = \text{Conv}(\text{Conv}(\hat{X}_i)) \quad (8)$$

Global Spatial. In order to establish long-distance dependence effectively, we introduce global spatial attention, which can be regarded as a supplement to local attention. Firstly, the feature $\hat{X}_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ is passed through Convolutional layer ($\mathbb{R}^{H_i \times W_i \times C_i} \rightarrow \mathbb{R}^{H_i \times W_i \times 1}$), Reshape and Softmax to obtain attention weight $\text{Att}_{gs} \in \mathbb{R}^{1 \times H_i \times W_i}$. Meanwhile, we employ the Reshape operation on original features to obtain the dimensionally transformed feature $R(\hat{X}_i) \in \mathbb{R}^{H_i \times W_i \times C_i}$. By this way, given the feature $R(\hat{X}_i)$ and the attention weight Att_{gs} , we can aggregate the features of all locations to form global context feature $\mathcal{P}_{gs} \in \mathbb{R}^{1 \times 1 \times C_i}$.

$$\text{Att}_{gs} = \text{Softmax}(\text{Conv}(R(\hat{X}_i))) \quad (9)$$

$$\mathcal{P}_{gs} = R(\hat{X}_i) \cdot \text{Att}_{gs} \quad (10)$$

Then, we further capture the dependency between feature channels through the multi-layer perceptron. Finally, we propagate the aggregated contextual features to all positions by an element-wise addition operation to generate global spatial information $\mathcal{F}_{gs} \in \mathbb{R}^{H_i \times W_i \times C_i}$.

$$\mathcal{F}_{gs} = \text{Conv}(\text{LN}(\text{Conv}(\mathcal{P}_{gs}))) + \hat{X}_i \quad (11)$$

where $\text{LN}(\cdot)$ and $R(\cdot)$ stand for LayerNorm and Reshape operation respectively. So the final gated feature is calculated as:

$$\mathcal{F}_i = \hat{X}_i \cdot \delta(\mathcal{F}_c + \mathcal{F}_{ls} + \mathcal{F}_{gs}) \quad (12)$$

Finally, we aggregate the two branches by element-wise addition operation and refine the output by a 3×3 convolution layer. Therefore, we can obtain the final refined feature $\mathcal{T}_i$.

3.3. Local feature aggregation module

In order to reduce the amount of computation, the two-stage detector applies the spatial pooling (e.g., RoI Pooling or RoI Align) layer to extract the single-scale features of each instance. However, the power of single-scale feature is usually constrained by insufficient contextual information. In this paper, we propose a Local Feature Aggregation Module (LFAM) to adaptively generate sufficient contexts for each instance, where different pooling kernels are utilized to aggregate multi-scale spatial-wise features. LFAM aims at optimizing the detection network from the local perspective.

As shown in Fig. 5, assume that the input feature is $\mathcal{T}_i \in \mathbb{R}^{H_i \times W_i \times C_i}$. Firstly, we use parallel pooling kernels with different sizes to extract

instance features $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_n\}$, where $\mathcal{R}_n \in \mathbb{R}^{\lambda_n \times \lambda_n \times c}$, and $\lambda_n < \lambda_{n-1}$. Therefore, the entire module is divided into multiple branches with different scales, i.e., high-resolution branch, medium-resolution branch, and low-resolution branch. In order to fuse these instance features, we use bilinear interpolation operation to upsample the low-resolution features and generate the fused global feature $\mathcal{M}$ by concatenate operation along the channel dimension:

$$\mathcal{R}'_i = \begin{cases} \mathcal{R}_i, & i = 1 \\ \text{Up}(\mathcal{R}_i), & i > 1 \end{cases} \quad (13)$$

$$\mathcal{M} = [\mathcal{R}'_1, \mathcal{R}'_2, \dots, \mathcal{R}'_n] \quad (14)$$

Then, we perform a channel reduction operation on the obtained global feature $\mathcal{M}$ to reduce parameters, and further split the global features to learn the weights $\mathcal{H}$ of different scales adaptively:

$$\mathcal{H} = \text{Conv}(\text{ReLU}(\text{Conv}(\mathcal{M}))) \quad (15)$$

Finally, we obtain the final refined instance feature map $\mathcal{I}_o$ by calculating the inner product of the weights $\mathcal{H}$ and the region features $\mathcal{R}'$: $\mathcal{I}_o = \sum_{d=1}^n \delta(\mathcal{H}_d) \cdot \mathcal{R}'_d$. The final feature map $\mathcal{I}_o$ emphasizes the spatial context of the instance and contributes more to the detection if served to detectors.

4. Experiments

4.1. Dataset

To verify the effectiveness of the proposed method, we perform extensive experiments on the MS COCO2017 benchmark (Lin et al., 2014), which is one of the most challenging and popular benchmarks for object detection and instance segmentation. MS COCO2017 is split into three subsets covering 80 object classes. Following common practice, we train models on training sets (115K images) and report main results on test sets (20K images), which is obtained from the evaluation server. In addition, we conduct the ablation study on the validation sets (5K images).

4.2. Evaluation metrics

For performance evaluation, we apply the evaluation metrics of MS COCO (Lin et al., 2014) to evaluate model performance for object detection and instance mask. The Average Precision (AP) is the main evaluation metric, which is averaged under Intersection-over-Union (IoU) thresholds from 0.5 to 0.95 with 0.05 interval. AP_{50} and AP_{75} are calculated when the $\text{IoU} = 0.5$ and $\text{IoU} = 0.75$. In addition, we divide the objects into different sets according to the object scales for evaluating the detection performance under different scales. We report AP_S on small-sized objects (less than 32×32), AP_M on medium-sized objects (from 32×32 to 96×96), and AP_L on large-sized objects (greater than 96×96). Note that AP_{box} and AP_{mask} denote AP for bounding box and segmentation mask, respectively.

Table 1

Object detection box AP on COCO test_dev. ‘Sch.’ is short for the training schedule.

Method	Backbone	Sch.	AP _{box}	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster R-CNN	ResNet50	1×	37.8	58.9	41.0	22.0	40.9	46.9
	Ours	1×	40.5[+2.7]	62.3[+3.4]	44.2[+3.2]	23.6[+1.6]	43.3[+2.4]	51.3[+4.4]
	ResNet101	1×	39.7	60.7	43.2	22.5	42.9	49.9
	Ours	1×	42.2[+2.5]	63.8[+3.1]	46.0[+2.8]	24.5[+2.0]	45.4[+2.5]	54.1[+4.2]
Mask R-CNN	ResNet50	1×	38.5	59.4	41.9	22.1	41.5	48.0
	Ours	1×	41.3[+2.8]	62.8[+3.4]	45.1[+3.2]	23.7[+1.6]	44.1[+2.6]	52.5[+4.5]
	ResNet101	1×	40.4	61.2	44.1	23.1	43.5	50.8
	Ours	1×	42.8[+2.4]	64.0[+2.8]	46.9[+2.8]	24.5[+1.4]	45.9[+2.4]	54.7[+3.9]
Cascade R-CNN	ResNet50	1×	40.6	59.2	44.0	23.0	43.4	51.1
	Ours	1×	43.3[+2.7]	62.5[+3.3]	47.1[+3.1]	25.2[+2.2]	46.1[+2.7]	54.9[+3.8]
	ResNet101	1×	42.3	61.0	46.0	23.9	45.4	53.6
	Ours	1×	44.8[+2.5]	64.0[+3.0]	48.7[+2.7]	25.7[+1.8]	47.8[+2.4]	57.5[+3.9]
Cascade Mask	ResNet50	1×	41.4	59.7	45.2	23.3	44.1	52.7
	Ours	1×	44.1[+2.7]	63.1[+3.4]	48.1[+2.9]	25.3[+2.0]	46.8[+2.7]	56.7[+4.0]
	ResNet101	1×	43.2	61.5	47.2	24.4	46.0	55.0
	Ours	1×	45.4[+2.2]	64.2[+2.7]	49.4[+2.2]	25.8[+1.4]	48.2[+2.2]	58.5[+3.5]
	ResNext101	1×	45.6	64.5	49.8	27.0	48.8	57.6
	Ours	1×	47.8[+2.2]	66.8[+2.3]	52.2[+2.4]	28.4[+1.4]	50.7[+1.9]	61.1[+3.5]
	Swin-T	1×	48.5	67.8	52.6	29.6	51.2	61.9
	Ours	1×	49.2[+0.7]	68.7[+0.9]	53.3[+0.7]	30.4[+0.8]	51.9[+0.7]	63.7[+1.8]
HTC	ResNet50	1×	42.6	61.8	46.3	24.5	45.1	54.3
	Ours	1×	44.8[+2.2]	64.2[+2.4]	48.6[+2.3]	25.8[+1.3]	47.3[+2.2]	57.5[+3.2]
	ResNet101	20e	45.1	64.2	49.1	25.7	47.8	58.0
	Ours	20e	46.8[+1.7]	66.2[+2.0]	50.9[+1.8]	27.3[+1.6]	49.5[+1.7]	60.3[+2.3]

Table 2

Instance Segmentation mask AP on COCO test_dev. ‘Sch.’ is short for the training schedule.

Method	Backbone	Sch.	AP _{mask}	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Mask R-CNN	ResNet50	1×	34.9	56.4	37.2	18.9	37.6	45.2
	Ours	1×	37.4[+2.5]	59.8[+3.4]	40.2[+3.0]	20.3[+1.4]	40.0[+2.4]	49.1[+3.9]
	ResNet101	1×	36.5	58.3	38.9	19.6	39.2	47.8
	Ours	1×	38.5[+2.0]	61.0[+2.7]	41.3[+2.4]	20.9[+1.3]	41.3[+2.1]	51.0[+3.2]
Cascade Mask	ResNet50	1×	36.7	57.0	39.0	19.1	38.5	47.4
	Ours	1×	38.6[+1.9]	60.4[+3.4]	41.6[+2.6]	20.9[+1.8]	40.9[+2.4]	51.0[+3.6]
	ResNet101	1×	37.6	58.9	40.5	20.1	40.2	49.4
	Ours	1×	39.6[+2.0]	61.6[+2.7]	42.8[+2.3]	21.3[+1.2]	42.2[+2.0]	52.7[+3.3]
	ResNext101	1×	39.7	61.7	42.9	22.3	42.5	51.7
	Ours	1×	41.6[+1.9]	64.2[+2.5]	45.1[+2.2]	23.4[+1.1]	44.3[+1.8]	54.8[+3.1]
HTC	Swin-T	1×	42.2	65.2	45.6	24.1	44.7	56.1
	Ours	1×	42.9[+0.7]	66.2[+1.0]	46.3[+0.7]	24.7[+0.6]	45.3[+0.6]	57.7[+1.6]
	ResNet50	1×	37.9	59.2	40.9	20.5	40.2	50.0
	Ours	1×	39.6[+1.7]	61.6[+2.4]	42.7[+1.8]	21.4[+0.9]	41.8[+1.6]	52.5[+2.5]
	ResNet101	20e	39.8	61.6	43.1	21.5	42.2	53.0
	Ours	20e	41.2[+1.4]	63.6[+2.0]	44.5[+1.4]	22.6[+1.1]	43.6[+1.4]	54.9[+1.9]

Table 3

Semantic segmentation: Performance comparisons on Cityscapes validation set.

Method	Backbone	mIoU (%)	mAcc (%)	aAcc (%)
Semantic FPN	ResNet50	74.5	81.9	95.8
	Ours	77.2[+2.7]	85.3[+3.4]	96.0
	ResNet50+PointRend	76.5	84.1	96.0
	Ours	78.0[+1.5]	85.7[+1.6]	96.1
	ResNet101	75.8	83.4	96.0
	Ours	78.2[+2.4]	85.8[+2.4]	96.1
	ResNet101+PointRend	78.3	85.7	96.2
	Ours	78.9[+0.6]	86.3[+0.6]	96.3
FCN	HRNet-Small	73.9	82.9	95.6
	Ours	75.5[+1.6]	84.2[+1.3]	95.9
	HRNet	77.2	85.3	96.2
	Ours	78.0[+0.8]	85.9[+0.6]	96.3

Table 4

Comparison with the state-of-the-art feature pyramid networks on COCO test_dev.

Method	Backbone	AP _{box}	#FLOPs	#Params
FPN (Lin, Dollár et al., 2017)	ResNet50	37.8	207.07	41.53
FPN (Lin, Dollár et al., 2017)	ResNet101	39.7	283.14	60.52
PAFPN (Liu et al., 2018)	ResNet50	38.0	368.13	53.86
Libra R-CNN (Pang et al., 2019)	ResNet50	38.7	208.12	41.79
CARAFE (Wang et al., 2019)	ResNet50	39.0	210.02	47.13
AugFPN (Guo et al., 2020)	ResNet50	38.8	220.21	43.58
ImFPN (Zhu et al., 2022)	ResNet50	39.5	N/A	N/A
DCIFPN (Xiao, Jiang, Li, & Gu, 2023)	ResNet50	38.8	N/A	N/A
MCFPN-Lite (Wang, Ji, Zhang, & Wu, 2023)	ResNet50	39.8	242.30	45.35
MCFPN (Wang et al., 2023)	ResNet50	40.6	494.30	57.88
AWBiFPN (Gao, Geng, Zhang, Wang, & Shao, 2024)	ResNet50	39.0	N/A	N/A
DyFPN (Zhu, Han, Yu, & Wang, 2020)	ResNet50	39.0	537.50	144.40
Ours	ResNet50	40.5	234.93	44.18

Table 5

Comparison with the state-of-the-art object detectors on COCO val2017.

Method	Backbone	Epochs	AP _{box}	AP _{s0}	AP ₇₅
RetinaNet (Lin, Goyal et al., 2017)	ResNet50	12	36.5	55.4	39.1
FCOS (Tian et al., 2019)	ResNet50	12	36.6	56.0	38.8
HTC (Chen, Pang et al., 2019)	ResNet50	12	42.6	61.8	46.3
DETR (Carion et al., 2020)	ResNet50	500	42.0	62.4	44.2
DETR+DC5 (Carion et al., 2020)	ResNet50	500	43.3	63.1	45.9
Sparse R-CNN (Sun et al., 2021)	ResNet50	12	37.9	56.0	40.5
Sparse R-CNN (Sun et al., 2021)	ResNet50	36	42.8	61.7	46.2
Deformable DETR (Zhu et al., 2021)	ResNet50	50	43.8	62.6	47.7
DN-DETR (Li et al., 2022)	ResNet50	12	38.6	N/A	N/A
Group DN-DETR (Chen, Chen et al., 2023)	ResNet50	12	40.6	N/A	N/A
SD-DETR (Zhang, Song, Liu, & Li, 2023)	ResNet50	12	39.5	59.3	41.7
SD-DETR-DC5 (Zhang et al., 2023)	ResNet50	12	40.6	59.9	43.1
HTC(Ours)	ResNet50	12	44.5	63.6	48.4

4.3. Implementation details

In this paper, we implement the proposed method based on the open-source toolbox MMDetection (Chen, Wang et al., 2019) by PyTorch. All experiments are performed on the workstation with 8 Nvidia RTX 2080Ti GPU cards. Following the common protocol, the input images are limited to 1333×800 pixels while keeping the original aspect ratio. In the training stage, we utilize the Stochastic Gradient Descent (SGD) algorithm to optimize models with a batch size of 16, which includes two training strategies (1x and 20e). The momentum is set to be 0.9 and the weight decay is set to be 0.0001. By default, models are trained for 12 epochs, which is called 1x training scheme. The learning rate starts from 0.02 and decreases by a factor of 0.1 after the 8th and 11th epoch. For the 20e training scheme, the learning rate is adjusted after the 16th and 19th epoch. Unless otherwise specified, all other hyper-parameters in this paper follow the settings in MMDetection.

4.4. Main results

Object Detection. In order to verify the effectiveness of our method, we construct variants with different detectors, backbones, training schemes and report the performance on COCO test_dev. For a fair comparison, we re-implement all baseline methods. As shown in Table 1, our method significantly outperforms all baseline methods by a large margin. By combining our method, Faster+ResNet50+FPN achieves 40.5% detection AP, which brings the increment of 2.7% over the baseline. When using ResNet101 as the backbone, our method achieves the 2.5% detection improvements. For Mask R-CNN, our method leads to gains of 2.8% and 2.4% for ResNet50 and ResNet101. Moreover, Cascade Mask R-CNN can increase by 2.7%, 2.2%, and 2.2% when using ResNet50, ResNet101, and ResNext101. Our methods further boost the performance of HTC by 2.2% and 1.7% AP gain for ResNet50 and ResNet101. Besides, we report comparative experimental results with mainstream feature pyramid networks (such as FPN, PAFPN, and DyFPN). As shown in the Table 4, compared with FPN, PAFPN,

and DyFPN, our method gains performance improvements of 2.7%, 2.5%, and 1.5% points AP_{box} respectively. Due to inherent limitation of the unidirectional information flow, traditional FPN has the lowest accuracy. On this basis, PAFPN achieves slightly better accuracy through bidirectional information flow. DyFPN adaptively executes the combination of convolutional layers through learnable gates. But both PAFPN and DyFPN directly use large convolutional kernel to process high-resolution feature map, thus hindering the efficiency. Our method adopts the low-resolution branch to supplement the original high-resolution branch from both global and local perspectives, enabling the balance between accuracy and efficiency as much as possible. FPN+ResNet101 outperforms FPN+ResNet50 due to its powerful capability of feature extraction. Notably, compared with FPN+ResNet101, our method boosts the ResNet50 architecture to achieve superior performance while having fewer parameters and computational complexity. Finally, we also compare the proposed method with the SOTA detectors. As shown in Table 5, our method achieves the best performance (44.5%), surpassing the competitive Deformable DETR by 0.7%. The intuitive visual comparison is shown in Fig. 6, which substantiates the fact that MSNet can accurately locate and classify objects of different sizes. In particular, MSNet can distinguish the dog with blurred appearance (first column) and the severely occluded cow (second column). We attribute this to the global-to-local progressive enhancement strategy adopted by MSNet, which can effectively obtain discriminative mixed-scale semantics.

Instance Segmentation. We evaluate our method on the instance segmentation task, see Table 2. For Mask R-CNN, our method can bring 2.5% and 2.0% AP for ResNet50 and ResNet101 respectively. When using the Cascade Mask R-CNN, the baseline method is increased by 1.9%, 2.0%, and 1.9% AP for ResNet50, ResNet101, and ResNext101. Meanwhile, our method boosts the powerful HTC by 1.7% and 1.4% AP gains for ResNet50, ResNet101. More recently, the transformer-based backbone network has brought remarkable improvement to object detection. Therefore, we construct the counterpart to verify the performance of MSNet. As shown in Table 1, paired with the backbone

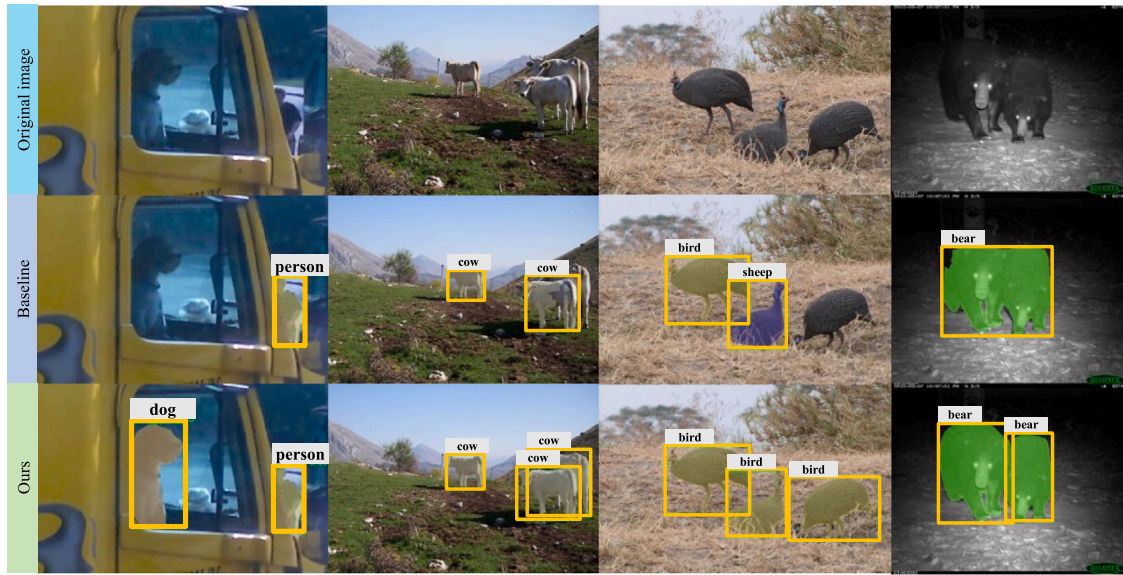


Fig. 6. Object detection and instance segmentation: Visual comparison between baseline and the proposed MSNet. The first row indicates the original sample, the second row showcases the result of Faster R-CNN, and the last row provides the result of MSNet.

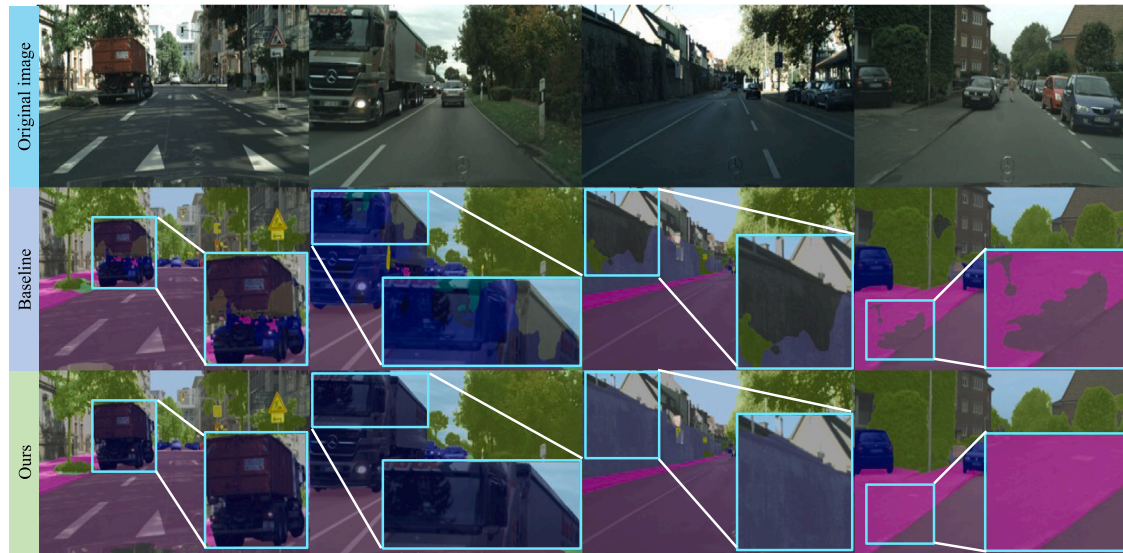


Fig. 7. Semantic segmentation: Visual comparison between baseline and the proposed MSNet. The first row indicates the original sample, the second row showcases the result of Semantic FPN, and the last row provides the result of MSNet.

of Swin-T, MSNet achieves 0.7%/0.7% performance improvements on AP_{box}/AP_{mask} compared with the baseline. As visualized in Fig. 6, MSNet shows obvious advantages over the baseline and achieves more accurate segmentation on object boundaries.

Semantic segmentation. To further verify the effectiveness of the proposed MSNet method on the semantic segmentation task, we also conduct extensive experiments based on the Cityscapes dataset (Cordts et al., 2016).

The Cityscapes is a large-scale dataset for semantic understanding of urban street scenes. It is formed by street scenes from 50 different cities, and has 5000 high-quality pixel annotation images of driving scenes in the urban environment. The Cityscapes is split into the training set, verification set, and test set, which include 2975 images, 500 images, and 1525 images respectively. We adopt the mean Intersection-over-Union (mIoU), mean accuracy of each class (mAcc), and all pixel accuracy (aAcc) as the primary evaluation metrics.

As presented in Table 3, our method gains consistent improvement by equipping with different backbones over the baseline method.

Specifically, paired with ResNet50, MSNet can boost the mIoU, mAcc, and aAcc by 2.7% points, 3.4% points and 0.2% points on Cityscapes dataset. Further, replacing with the stronger backbone (ResNet101), MSNet achieves the performance gains of 2.4% points, 2.4% points and 0.1% points on mIoU, mAcc, and aAcc, respectively.

Meanwhile, our method is still suitable for the network architectures with sophisticated mask heads, e.g. PointRend (Kirillov, Wu, He, & Girshick, 2020). Like PointRend+ResNet50, MSNet improves the performance by 1.5% points mIoU, 1.6% points mAcc and 0.1% points aAcc compared with Semantic FPN. Fig. 7 visualizes the semantic segmentation results on the validation set of Cityscapes. It is clearly seen that MSNet yields better results compared with baseline (Semantic FPN) and provides more consistent labels on large instances.

4.5. Ablation studies

In this sub-section, we conduct comprehensive ablation studies to analyze the effectiveness of components in our method on COCO

Table 6

Effectiveness of each component on COCO val2017. GFAM: Global Feature Aggregation module. GFEM: Global Feature Enhancement module. LFAM: Local Feature Aggregation module.

GFAM	GFEM	LFAM	#FLOPS	#Params	AP _{box}	AP ₅₀
			207.07	41.53	37.4	58.1
✓			228.25	41.53	39.3	60.5
	✓		208.95	43.97	38.9	60.3
		✓	212.03	41.73	38.0	58.9
✓	✓		230.14	43.97	39.9	61.2
	✓	✓	213.87	44.18	39.3	60.7
✓		✓	232.95	41.73	39.5	60.6
✓	✓	✓	234.93	44.18	40.3	61.8

Table 7

Analysis of the GFAM ratio k on COCO val2017.

k	#FLOPS	#Params	AP _{box}	AP ₅₀	AP ₇₅
0.3	221.87	44.18	40.0	61.3	43.7
0.4	227.50	44.18	40.0	61.3	43.8
0.5	234.93	44.18	40.3	61.8	44.2
0.6	244.13	44.18	40.3	61.6	43.8
0.7	255.22	44.18	40.2	61.7	43.5

Table 8

Analysis of the GFAM pooling operation on COCO val2017.

Setting	AP	AP ₅₀	AP ₇₅
Max	39.9	61.3	43.4
Avg	40.1	61.6	43.6
Max+Avg	40.3	61.8	44.2

Table 9

Analysis of the GFAM convolution kernel on COCO val2017.

Setting	AP	AP ₅₀	AP ₇₅
3	40.0	61.4	43.5
5	40.1	61.6	43.4
7	40.3	61.8	44.2
9	40.2	61.8	43.8

val2017. Note that all experiments adopt Faster+ResNet50+FPN as the default baseline method for fair comparison.

Effectiveness of component. In order to analyze the effectiveness of individual components in our proposed method, we calculate the detection performance of variants, *i.e.*, a combination of different components. As shown in Table 6, each component contributes to superior performance. Specifically, the GFAM module improves the baseline method by 1.9% AP. We suppose that the features extracted from low-res images effectively enrich the original features at the different stages. When incorporating the GFEM module, the detection performance is further improved from 37.4% to 38.9%. We attribute these enhancements to the fact that the global feature enhancement method can effectively suppress the information loss of the highest-level feature by downsampling its original features. Furthermore, LFAM achieves 0.6% detection AP gains, which confirms the benefit of multi-scale align kernel. We show that when more components are combined, the effect is more obvious. For example, the combination of GFAM and GFEM brings 2.5% AP gains over the baseline. When the three modules work together, the gains of AP up to 2.9% further.

Analysis of Global Feature Aggregation. In order to further demonstrate the robustness of the GFAM against hyper-parameters, we firstly vary the downsampling ratio k to analyze its effect on the detection performance. The results are reported in Table 7. The model reaches its best performance when the ratio is 0.5. We speculate there is limited information in an image when $k \leq 0.4$, which is not enough to enrich the high-res features. However, if we set $k \geq 0.6$, small difference in size may not explicitly form supplementary information in the feature space. Considering the performance and parameters, $k = 0.5$ is taken

Table 10

Ablation study of the GFEM on COCO val2017.

Setting	AP _{box}	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Baseline	37.4	58.1	40.4	21.2	41	48.1
+ Low-res	38.1	59.3	41.3	22.4	41.3	49.7
+ MFEB	38.9	60.3	42.2	22.7	42.4	50.2
Channel	38.5	59.8	41.8	22.3	42	49.6
Local Spatial	38.3	59.4	41.2	22.6	41.4	49.8
Global Spatial	38.5	59.9	41.7	22.4	41.8	49.7
All	38.9	60.3	42.2	22.7	42.4	50.2

Table 11

Analysis of the GFEM ratio d on COCO val2017.

d	#FLOPS	#Params	AP _{box}	AP ₅₀	AP ₇₅
0.3	208.20	43.97	39.9	61.4	43.9
0.4	208.57	43.97	40.1	61.6	44.0
0.5	208.95	43.97	40.3	61.8	44.2
0.6	209.33	43.97	40.3	61.9	43.7
0.7	209.71	43.97	40.3	61.7	43.8

Table 12

Object detection performance of the proposed GFEM and existing attention mechanisms on COCO val2017.

Config	AP	AP ₅₀	AP ₇₅	#Params	#FLOPS
Baseline	37.4	58.4	40.6	41.53	207.07
PPM	38.2	59.5	41.3	42.64	207.92
ASPP	38.4	59.8	41.4	44.28	209.75
Non-Local	38.2	59.7	41.5	42.32	207.85
Ours	38.9	60.3	42.2	43.97	208.95

Table 13

Analysis of the LFAM pooling kernel on COCO val2017.

Setting	AP _{box}	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Baseline	37.4	58.1	40.4	21.2	41.0	48.1
7 + 5	37.8	58.6	40.8	22.0	41.6	48.7
7 + 5 + 3	38.0	58.9	41.3	22.2	41.4	49.3
9 + 7 + 5	38.0	59.0	41.3	22.6	41.5	49.1

as default for all the other experiments. Experimental results also demonstrate that the original feature of the backbone network can be effectively enriched by introducing the low-res image.

In addition, we analyze the effect of different pooling operations in GFAM. In order to reduce the amount of parameters, we choose two typical pooling operations, namely average pooling and global pooling. As shown in Table 8, the combination scheme performs better than the single scheme. This may be due to the fact that two pooling operations can benefit each other when combined together.

Finally, as shown in Table 9, we also analyze the impact of convolution kernel in GFAM on model performance. From the results, we can see that the performance first increases and then stabilizes as the convolution kernel increases. Since both the channel dimension of input and output are 2, the amount of parameters brought by this convolution kernel is negligible. So, we set the kernel size of convolution layer to 7 in this paper.

Insight of Global Feature Enhancement. To lend insight into the GFEM module further, we perform numerous ablative studies whose results are listed in Table 10. As for the metric of AP_{box}, GFEM enriches the original high-res branch via the low-res branch, yielding an improvement of 0.7% points. When equipped with MFEB, the detection performance is improved from 38.1% to 38.9%. Moreover, we probe into sub-branches of MFEB and find that the schema of MFEB brings performance improvements of 0.4%, 0.2% and 0.4% AP in terms of channel, local spatial and global spatial, respectively. These improvements benefit from the design of MFEB where more context information can be excavated from multi-level branches.

In Table 11, we also show the effects of different downsampling ratios in the proposed GFEM. It can be seen from the results that the best performance appears when the ratio is equal to 0.5. And, it also achieves a good balance between efficiency and performance. Higher ratio fails to introduce scale-specific semantic information, while lower ratio results in insufficient information supplementation.

Considering that there are a variety of attention mechanisms, we also conduct corresponding experiments to verify the effectiveness of the proposed GFEM. We compare three mainstream attention mechanisms and the results are listed in Table 12. Compared with PPM (Zhao, Shi, Qi, Wang, & Jia, 2017), ASPP (Chen, Papandreou, Kokkinos, Murphy, & Yuille, 2018), Non-Local (Wang et al., 2018), we can see that the proposed GFEM achieves the best performance. Compared with existing methods, GFEM adopts a parallel multi-branch structure to learn scale-specific information, and further uses MFEB to extract multi-granularity information, thereby suppressing interference information from the background and irrelevant contents.

Influence of Local Feature Aggregation. To investigate the influence of the pooling kernel in LFAM on the detection performance, we construct the variants of MSNet via different kernel sizes, and show the experimental results in Table 13. According to the common settings, the default kernel size in the detector is $\{7 \times 7\}$. From Table 13, we can see that the proposed method achieves the best performance when the instance sizes are $\{7 \times 7, 5 \times 5, 3 \times 3\}$ and $\{14 \times 14, 7 \times 7, 5 \times 5\}$. Considering a decent trade-off between efficiency and accuracy, we take $\{7 \times 7, 5 \times 5, 3 \times 3\}$ as the default setting in the LFAM for all experiments. Therefore, the ablative results further demonstrate the rationale of multiple kernel size protocol instead of a default $\{7 \times 7\}$ instance size in the baseline method. The single fixed pooling kernel contains limited contextual information and cannot meet the needs of different objects. However, the proposed method can flexibly extract sufficient contextual information, avoiding the impact of single fixed pooling kernel.

5. Conclusion

In this paper, we propose a simple yet efficient Mixed-Scale Network (MSNet), which is inspired by the human behavior of zooming blurred images. Specifically, the global feature aggregation module aligns features from different resolutions in an adaptive manner and the global feature enhancement module suppresses the interference context from the background and provides more relevant context information for high-res features. Meanwhile, the local feature aggregation module utilizes multi-scale pooling kernels to adaptively instill diverse spatial-wise contexts into each instance. Extensive experimental results show the superiority of MSNet on numerous excellent detectors. Additionally, the extension of MSNet on the semantic segmentation task substantiates the effectiveness of the proposed method, where MSNet performs favorably against the baseline methods. As a plug-and-play module, we believe that MSNet can provide a new baseline for a variety of visual tasks. For future works, we will explore a more effective and efficient network to achieve better performance in terms of both accuracy and speed.

CRedit authorship contribution statement

Boying Wang: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Ruyi Ji:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Libo Zhang:** Visualization, Funding acquisition, Writing – review & editing. **Yanjun Wu:** Formal analysis, Funding acquisition, Writing – review & editing. **Jing Liu:** Validation, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Youth Project of Applied Basic Research Project of Shanxi Province (entitled "Research on General Object Detection Algorithms Driven by Multi-Source Imbalanced Data"), the Foundation of Shanxi Key Laboratory of Machine Vision and Virtual Reality (No. 447-110103), the National Natural Science Foundation of China (No. 62306314), and the Key Research Program of Frontier Sciences, CAS, Grant No. ZDBS-LY-JSC038. Libo Zhang was supported CAMS Innovation Fund for Medical Sciences (No. 2021-I2M-C&T-A-011), CAAI-Huawei MindSpore Open Fund and Youth Innovation Promotion Association, CAS (2020111).

Data availability

The data that has been used is confidential.

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